Khwaja Abdul Samad

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.66
Saiyyid Hamid Sr. Sec. School, AMU, Aligarh	Class XII (AMU Board)	2019	79.40%
Saiyyid Hamid Sr. Sec. School, AMU, Aligarh	Class X (AMU Board)	2017	9.60%

PROJECTS

Dynamic Topic Modeling & Citation Prediction | *M.Tech Thesis* | *Prof. Srijith P. K.*

- Implemented and evaluated **DTM, D-ETM, and DSNTM** on the SciEvo scientific-literature corpus to model topic evolution across **7 yearly time slices**, using a common preprocessing and evaluation pipeline.
- Extended DSNTM with **citation-based regularization**, improving **NPMI coherence by 28.9%** and **topic quality by 32.0%** over the base DSNTM in 3-seed evaluation, while analyzing the trade-off with document-completion perplexity.
- Developing **HD-ETM**, a joint **D-ETM + Hawkes process** framework that uses dynamic topic representations to model and predict **citation rates and timing**, with a topic-space excitation mechanism for scalable citation-history modeling.

Multi-Agent IPO & DRHP Analysis System | *GitHub*

- Built a **LangGraph multi-agent system** that analyzes 300-550 page IPO offer documents using five specialized agents for risk factors, financial health, business & promoters, offer structure, and legal & approvals feeding into a final synthesis agent that produces a scored, page-cited investment report.
- Designed for **auditable RAG** using hybrid **FAISS + BM25 retrieval with Reciprocal Rank Fusion**, section-aware filtering, and page-level citations; arithmetic, financial ratios, and scoring are computed deterministically in Python rather than by the LLM, validated across **39 hand-checked facts from four offer documents**.
- Developed the full-stack system with **FastAPI, Next.js/TypeScript, PyMuPDF, Sentence Transformers, and Groq**, with background job processing, per-agent caching, and a token-rate limiter for reliable inference under API limits.

Business Entity Resolution | *Amazon ML Challenge 2026* **|** *GitHub*

- Built a **retrieve-then-rank entity resolution pipeline** for matching ~**1.7M businesses against ~10M noisy records**, generating **88M candidate pairs** using multi-view TF-IDF, phonetic search, and **LaBSE embeddings** for multilingual names across Indian scripts.
- Trained a **two-stage LightGBM matcher** with similarity, address/house-number edit patterns, typo-vs-word-swap signals, and candidate-competition features, followed by exclusive assignment of each record to at most one business.
- Achieved **0.989 macro F0.5** on a validation set mirroring the test set's distractor density and **0.980 on the leaderboard**, with candidate-rescue search and label-free normalization improving generalization to **France, a country absent from the training data**.

TransactiWar - Secure Banking Web Application | *Network Security* | *GitHub*

- **Developed as part of a 5-member team** a secure, framework-free banking application using PHP and PostgreSQL, with authentication, session security, CSRF protection, rate limiting, and activity logging implemented from scratch.
- Engineered **race-condition-safe money transfers** using PostgreSQL transactions, row-level locking, balance validation, and deadlock-safe lock ordering to maintain consistent balances under concurrent requests.
- Implemented defenses against **SQL injection, XSS, brute-force attacks, IDOR, and malicious file uploads** using prepared statements, input validation, BCrypt hashing, and GD-based image re-encoding; deployed with HTTPS/TLS with CSP, HSTS, and security headers using Docker.

SKILLS

Programming Languages: C, C++, Python, JavaScript, SQL

ML/DL: PyTorch, Scikit-learn, NumPy, Pandas, CNNs, Transformers

LLM/GenAI: RAG, Agentic RAG, LangChain, LangGraph, Sentence Transformers, LLMs

Backend/Database: FastAPI, REST APIs, MySQL, ChromaDB, FAISS

Web/Tools: Next.js, React, Tailwind CSS, Docker, Kubernetes, Git, GitHub, Linux, Wireshark, ns-3

RELEVANT COURSES

Foundations of Machine Learning | Deep Learning | Fraud Analytics | Advanced Data Structures and Algorithms

Advanced Computer Networks | Network Security | Networked Wireless Systems

POSITIONS OF RESPONSIBILITY

Teaching Assistant | IIT Hyderabad

- Data Structures & Applications -- July 2025 - Nov 2025
- Compilers-I -- March 2026 - April 2026
- Foundations of Machine Learning -- July 2026 - Nov 2026

ACHIEVEMENTS

- **LeetCode: Knight,** solved **2050+ problems**; maximum contest rating **2035**; ranked **358th/35.7K** in Weekly Contest 400 and **510th/21.4K** in Biweekly Contest 116
- **CodeChef: 3-Star**, maximum rating **1637**; ranked **109th/3.7K** in Starters 92 Div. 3
- **Codeforces: Pupil**, maximum rating **1393**
- **GeeksforGeeks:** solved **1000+ problems**

EXTRACURRICULAR

Table Tennis | Badminton | BGMI | Word Puzzles *(Wordle, Strands, Minute Cryptic)*